Baorian Nuchged
baorian@utexas.edu | https://boroooo.github.io/
Linguistics PhD student

Research Interests

Speech processing, neurolinguistics, phonetics, and information-theoretic approaches to language.

Education

University of Texas at Austin Aug 2023 - present
Linguistics - Neurolinguistics & Phonetics, PhD Program Austin, Texas

Beijing Language and Culture University Sep 2020 - Jul 2023
Language Intelligence and Technology Master (with merit) Beijing, China
Thesis: Quantifying information loss in second language speech using information theory

East China Normal University Aug 2015 - May 2019
English Bachelor Shanghai
Thesis: A Phonological Study on Chinese Loanwords in Mongolian (Outstanding Graduation Dissertation Award)

Research Experience

Graduate Research Assistant at the University of Texas at Austin May 2026 – Aug 2026
Advised by Dr. Megan Crowhurst
I'm working on the acoustic effects on rhythmic grouping of different languages.

Graduate Student Researcher at Neurolinguistics Lab at the University of Texas at Austin Aug 2023 - Present
Advised by Dr Fernando Llanos
I'm working on the mechanisms of auditory information processing in the human brain, focusing on uncovering the compressed sensing processes and examining how they differ from those found in machine processing such as uniform sampling and the Nyquist theorem.

Graduate Research Assistant at the University of Texas at Austin Aug 2024 – Dec 2025
Advised by Dr. Scott Myers
I am working on the acoustic analysis of Luganda H-tone syllables, to find the f0 timing and its relationship with the syllable structure.

Research Assistant at Speech Acquisition and Intelligent Technology Lab (SAIT) Aug 2020 - Jun 2023
Dr. Zhang's lab studies the speech acquisition of second-language learners and hearing-impaired children. I mainly worked on the human perception of second-language speech.
-- Responsibilities include conducting perception experiments, speech data annotation, processing, and management.

Research Assistant at the Department of Language and Literature of Ethnic Minorities of China Jun 2021 - Sep 2021
Prof. Bao's department focuses on Mongolian phonetics and phonology.
-- Responsibilities include data analysis, dialect speech recording, and academic paper writing.

Funded Project: A Phonological Study on Chinese Loanwords in Mongolian Sep 2018 - Jun 2019
[Undergrad University]
Student project funded by ECNU undergraduate research funding
-- Corpus analysis on Mongolian Loanword, completed by myself.

Teaching Experience

Fall 2026 Assistant Instructor
Introduction to the Study of Language
Department of Linguistics, University of Texas at Austin

Spring 2026 Guest Lecturer
Phonetics: Production and Perception of Speech Sound
Department of Linguistics, University of Texas at Austin

Spring 2026 Teaching Assistant
Phonetics: Production and Perception of Speech Sound (Instructor: Scott Myers)
Department of Linguistics, University of Texas at Austin

Fall 2024 Guest Lecturer
Introduction to the Study of Language
Department of Linguistics, University of Texas at Austin

Mentoring Experience

Spring, Fall 2026 Neurolinguistcis lab undergraduate mentoring
EEG experiment setup

Fall 2026 Undergraduate mentoring on NSF project: Effects of Complex Speech Sound Systems on Language Dependent Auditory Plasticity.
Speech annotation

Publications and Presentations

- **Baorian Nuchged**, Christian Stilp, Keith Kluender, and Fernando Llanos. “Changes in Narrowband Spectral Energy Predict Cortical Encoding of Continuous Speech.” *Under review*.
- **Baorian Nuchged** and Adiya Mingaad. “Predicting Mongolian Unstable n: Lexical, Phonotactic, and Morphosyntactic Factors in the Dative-Locative.” *Under review.*
- Scott Myers, Anatole Kiriggwajjo, **Baorian Nuchged**, Saudah Namyalo. “Segmentation and gestural timing in Luganda consonant-glide sequences. ” *Under review.*
- Linkai Peng and **Baorian Nuchged**. “Represented but Ignored: A Causal Account of Prosodic Underuse in Audio-Language Models.” arXiv preprint arXiv:2608.19211, 2026.
- **Baorian Nuchged**, Christian Stilp, Keith Kluender, and Fernando Llanos. “Cochlea-Scaled Entropy Predicts Robust Neural Coding of Speech Envelopes.” *Seventeenth Annual Meeting of the Society for the Neurobiology of Language (SNL)*, 12–14 Sept. 2025, Gallaudet University, Washington, D.C. Poster.
- **Baorian Nuchged** and Rajka Smiljanic. “Protective Face Masks Affect Clearly Produced Diphthongs by L1 and L2 English Talkers.” *The Journal of the Acoustical Society of America* 155.3 Supplement (2024): A315.
- **Rian Bao**, Linkai Peng, Yuchen Yan, and Jinsong Zhang. “An Exploratory Study for Quantifying the Contextual Information for Successful Chinese L2 Speech Comprehension.” *The 13th International Symposium on Chinese Spoken Language Processing (ISCSLP)*, 2022. DOI: 10.1109/ISCSLP57327.2022.10038037.
- Yingming Gao, **Baorian Nuchged**, Ya Li, and Linkai Peng. “An Investigation of Applying Large Language Models to Spoken Language Learning.” *Applied Sciences* 14.1 (2024): 224. DOI: 10.3390/app14010224.

- **Rian Bao**, Linkai Peng, Yingming Gao, and Jinsong Zhang. “The Contribution of Phonological and Fluency Factors to Chinese L2 Comprehensibility Ratings: A Case Study of Urdu-Speaking Learners.” *The 13th International Symposium on Chinese Spoken Language Processing (ISCSLP)*, 2022. DOI: 10.1109/ISCSLP57327.2022.10038233.
- **Rian Bao**, Linkai Peng, and Jinsong Zhang. "The contributions of initials and finals to L2 Chinese comprehensibility based on functional load principle." 2022 International Conference on Asian Language Processing (IALP). IEEE, 2022.
- **Rian Bao**, et al. "The Importance of Lexical Tone for Sentence Understanding: Utilizing Functional Load Principle to Simulate Comprehension Process." 2022 International Conference on Asian Language Processing (IALP). IEEE, 2022.
- Bao Yin, and **Rian Bao**. "Contextual Vowel Nasalization in Mongolian." 2021 International Conference on Asian Language Processing (IALP). IEEE, 2021.

HONORS & AWARDS

UT Austin Professional Development Award ($1200)	2024, 2025
Graduate Student Fellowship at University of Texas at Austin ($25,000)	Aug 2023
Outstanding Master's Thesis Award of Beijing Language and Culture University	Jun 2023
Merit Student of Beijing Language and Culture University	Dec 2021
Best Student Paper Award in Phonetics Conference of China	Jul 2021
BLCU graduate student scholarship (¥6,000 CNY yearly)	Sep 2020, Sep 2021 and Sep 2022
BLCU graduate student research fund (¥2,000 CNY)	2022
Outstanding Bachelor's Thesis Award of East China Normal University	Jul 2019
ECNU undergraduate research fund (¥2,000 CNY)	2018
Thesis: A Phonological Study on Chinese Loanwords in Mongolian (Outstanding Graduation Thesis Award)	

WORK EXPERIENCE

Nio — Feb 2023 - May 2023

Data Specialist (Intern) — Beijing

Organizing and analyzing the wakeup speech and text data of the voice assistant Nomi. I gained experience in data management from this internship.

ByteDance (Tiktok) — Mar 2022 - Jul 2022

English Text-to-Speech Linguistic program manager (Intern) — Beijing

Led the design and implementation of highly human-like Text-to-Speech (TTS) projects Ava and Henry (click to download demo).

Project link:
https://www.volcengine.com/docs/6561/97465#%E5%A4%9A%E8%AF%AD%E7%A7%8D

DataBaker — Oct 2021 - Dec 2021

Speech Data Annotator (Intern) — Beijing

ACADEMIC SERVICE

2024 Texas Linguistics Society Annual Conference	Conference coordinator
2025 Texas Linguistics Society Annual Conference	Conference organizing committee

SKILLS LIST

Programming Languages: Python scripting (Fluent), Praat scripting (Fluent), R scripting for statistical analysis, Matlab scripting, knowledge of the Linux operating system
Data Analysis: R (ggplot2, lm, lme4, olsrr)
Experimental Design: Finding Five, Gorilla
Writing and Communication: Microsoft Office, PowerPoint, Excel, LaTex
Languages: Mongolian (native), Chinese Mandarin (native), English (fluent), Spanish (entry-level)